

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

10 answers · Rationale-rich · Teach from doc

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Q1**C**

C. $16p + 37 = 165$

Choice C is correct. It's given that John made a \$ 16 payment each month for p months. The total amount of these payments can be represented by the expression $16p$. The down payment can be added to that amount to find the total amount John paid, yielding the expression $16p + 37$. It's given that John paid a total of \$ 165. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation $16p + 37 = 165$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D**

D. 30

Choice D is correct. It's given that when $x = 0$, $f_x = 30$. Substituting 0 for x and 30 for f_x in the given function yields $30 = 0 + b$, or $30 = b$. Therefore, the value of b is 30.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q3**A**

A. $3w + 5r = 14$

Choice A is correct. Since Jay walks at a speed of 3 miles per hour for w hours, Jay walks a total of $3w$ miles. Since Jay runs at a speed of 5 miles per hour for r hours, Jay runs a total of $5r$ miles. Therefore, the total number of miles Jay travels can be represented by $3w + 5r$. Since the combined total number of miles is 14, the equation $3w + 5r = 14$ represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q4**C**

C. 27

Choice C is correct. It's given that $x = 7$. Substituting 7 for x into the given expression $x + 20$ yields $7 + 20$, which is equivalent to 27.

Choice A is incorrect. This is the value of $x + 6$.

Choice B is incorrect. This is the value of $x + 13$.

Choice D is incorrect. This is the value of $x + 27$.

Q5**D****D.** 5

Choice D is correct. It's given that the function f is defined by $fx = \frac{1}{2}x + 6$. Substituting 4 for x in the given function yields $f4 = \frac{1}{2}4 + 6$, or $f4 = 5$. Therefore, the value of $f4$ is 5.

Choice A is incorrect. This is the value of $24 + 6$, not $\frac{1}{2}4 + 6$.

Choice B is incorrect. This is the value of $2 + 4 + 6$, not $\frac{1}{2}4 + 6$.

Choice C is incorrect. This is the value of $4 + 6$, not $\frac{1}{2}4 + 6$.

Q6**B****B.** $y = -\frac{1}{2}x + 3$

Choice B is correct. A line in the xy -plane with a slope of m and a y -intercept of $0, b$ can be represented by the equation $y = mx + b$. It's given that the line has a slope of $-\frac{1}{2}$. Therefore, $m = -\frac{1}{2}$. It's also given that the line passes through the point $0, 3$. Therefore, $b = 3$. Substituting $-\frac{1}{2}$ for m and 3 for b in the equation $y = mx + b$ yields $y = -\frac{1}{2}x + 3$. Therefore, the equation $y = -\frac{1}{2}x + 3$ represents this line.

Choice A is incorrect. This equation represents a line in the xy -plane that passes through the point $0, -3$, not $0, 3$.

Choice C is incorrect. This equation represents a line in the xy -plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$, and passes through the point $0, -3$, not $0, 3$.

Choice D is incorrect. This equation represents a line in the xy -plane that has a slope of $\frac{1}{2}$, not $-\frac{1}{2}$.

Q7**C****C. 8**

Choice C is correct. It's given that a gym charges its members a onetime \$ 36 enrollment fee and a membership fee of \$ 19 per month. Let x represent the number of months at the gym after which a member will have been charged a total of \$ 188. If there are no charges other than the enrollment fee and the membership fee, the equation $36 + 19x = 188$ can be used to represent this situation. Subtracting 36 from both sides of this equation yields $19x = 152$. Dividing both sides of this equation by 19 yields $x = 8$. Therefore, a member will have been charged a total of \$ 188 after 8 months at the gym.

Choice A is incorrect. A member will have been charged a total of $\$ 36 + 19 \times 4$, or \$ 112, after 4 months at the gym.

Choice B is incorrect. A member will have been charged a total of $\$ 36 + 19 \times 5$, or \$ 131, after 5 months at the gym.

Choice D is incorrect. A member will have been charged a total of $\$ 36 + 19 \times 10$, or \$ 226, after 10 months at the gym.

Q8**11**

The correct answer is 11. It's given that the function $f(x) = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when x games are played. Substituting 58 for $f(x)$ in the given equation yields $58 = 14 + 4x$. Subtracting 14 from each side of this equation yields $44 = 4x$. Dividing each side of this equation by 4 yields $11 = x$. Therefore, 11 games can be played for a total cost of \$ 58.

Q9**B**

B. $x - y = 201$

Choice B is correct. Of the city's total expense budget for one year, the city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. This means that the expression $y + 201$ represents the total expense budget, in millions of dollars, for one year. It's given that the total expense budget for one year is x million dollars. It follows then that the expression $y + 201$ is equivalent to x , or $y + 201 = x$. Subtracting y from both sides of this equation yields $201 = x - y$. By the symmetric property of equality, this is the same as $x - y = 201$.

Choices A and C are incorrect. Because it's given that the total expense budget for one year, x million dollars, is comprised of the departmental expenses, y million dollars, and all other expenses, 201 million dollars, the expressions $x + y$ and $2x - y$ both must be equivalent to a value greater than 201 million dollars. Therefore, the equations $x + y = 201$ and $2x - y = 201$ aren't true. Choice D is incorrect. The value of x must be greater than the value of y . Therefore, $y - x = 201$ can't represent this relationship.

Q10**D**

D. $8x + 3 = 83$

Choice D is correct. The given phrase "8 times a number x " can be represented by the expression $8x$. The given phrase "3 more than" indicates an increase of 3 to a quantity. Therefore "3 more than 8 times a number x " can be represented by the expression $8x + 3$. Since it's given that 3 more than 8 times a number x is equal to 83, it follows that $8x + 3$ is equal to 83, or $8x + 3 = 83$. Therefore, the equation that represents this situation is $8x + 3 = 83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83.

Choice B is incorrect. This equation represents 8 times a number x is equal to 3 more than 83.

Choice C is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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